Lecture 6: **NP**-completeness

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Polynomial-time Reductions

Problem X polynomially reduces to problem Y if an arbitrary instance of problem X can be:

- transformed to an instance of problem Y in a polynomial number of steps, and then
- solved using a polynomial number of calls to an oracle that solve problem Y.

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- solved using a polynomial number of calls to an oracle that solve problem Y.

Notation: X is polynomial-time reducible to Y is written $X \leq Y$,

If $X \leq Y$ and $Y \leq X$, then X and Y are equivalent
(with respect to polynomial reductions).

Example

Recall from Lecture 3:

Theorem

INDEPENDENT SET *is equivalent to* VERTEX COVER.

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As the problems are equivalent we also have:

$$\text{VERTEX COVER} \leq \text{INDEPENDENT SET}$$

NP-completeness proofs

To prove that a problem Π is **NP**-complete we now just have to perform two steps:

1. Show that Π belongs to **NP**
2. Find a known **NP**-complete problem Π' and show $\Pi' \leq \Pi$

Thousands of problems have now been shown to be **NP**-complete (for an early survey see Garey and Johnson).

Note: If we can complete Step 2 but not Step 1, then we say that Π is **NP**-*hard*

Three proof techniques

Three general methods to prove that an **NP**-complete problem Π' is polynomial-time reducible to a problem Π:

- Restriction
  - Show that Π' is a subproblem of Π
- Local replacement
  - Show that every basic unit in an instance of Π' can be replaced by a different structure in a uniform way to obtain an instance of Π.
- Component design
  - Show that the constituents of an instance of Π can be used to "design" components that can be combined to "realize" (or to "encode") instances of Π'.

NP-completeness of VERTEX COVER

VERTEX COVER
Instance: A graph $G = (V, E)$, and a natural number k.
Question: Is there a set $W \subseteq V$, with $|W| \leq k$, such that for each edge $(i, j) \in E$,

$$\{i, j\} \cap W \neq \emptyset$$

The problem VERTEX COVER is in **NP** (the list of vertices in the cover is the certificate)

To show that VERTEX COVER is **NP**-complete we shall reduce SATISFIABILITY to VERTEX COVER

NP-completeness of VERTEX COVER

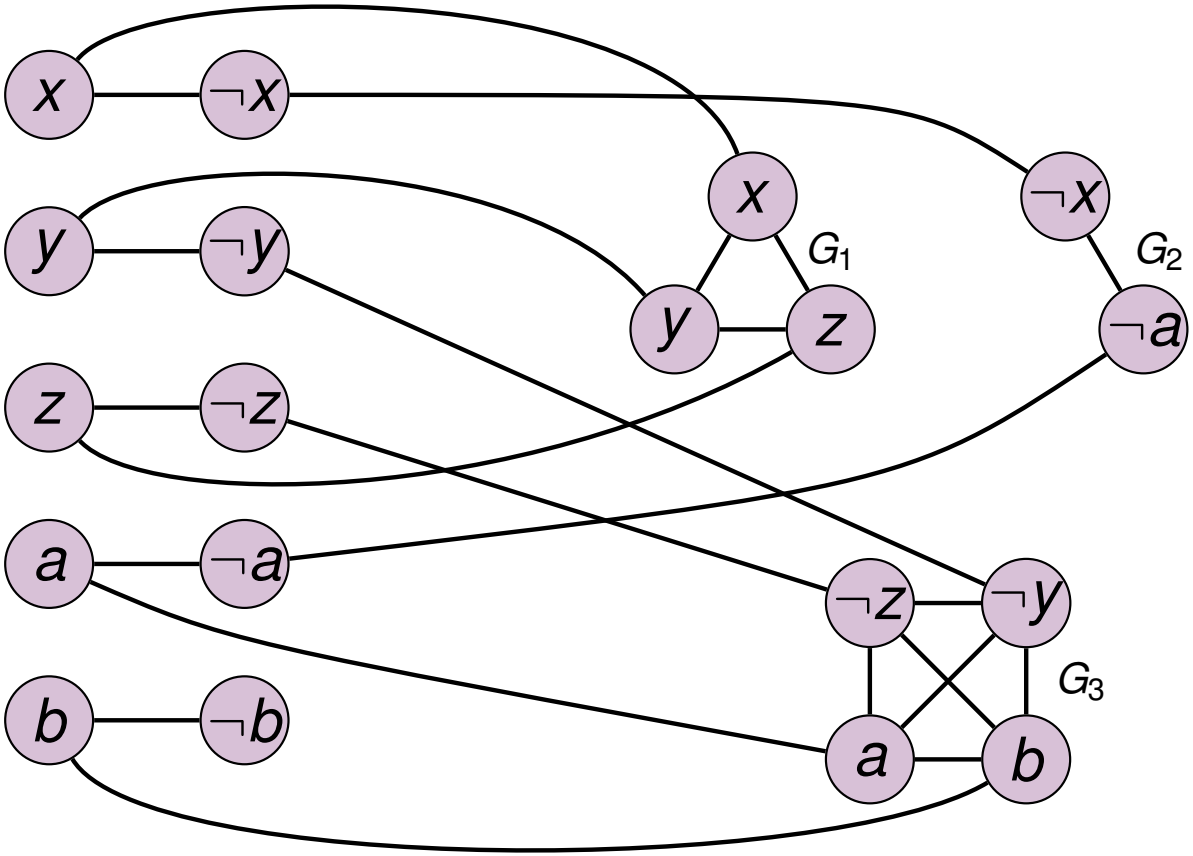
Given a formula f with n variables and clauses $C_1, C_2, \ldots, C_m$

- For each variable x create two adjacent vertices x^t and x^f to represent the *literals* x and $\neg x$
- For each clause C_j of size n_j create a complete subgraph G_j with vertices connected to corresponding literals
- Set $k = n + \sum_{j=1}^{m}(n_j - 1)$

This construction can be carried out in polynomial time

Example construction

$$\underbrace{(x \vee y \vee z)}_{C_1} \wedge \underbrace{(\neg x \vee \neg a)}_{C_2} \wedge \underbrace{(a \vee b \vee \neg y \vee \neg z)}_{C_3}$$



$k = 5 + 2 + 1 + 3 = 11$

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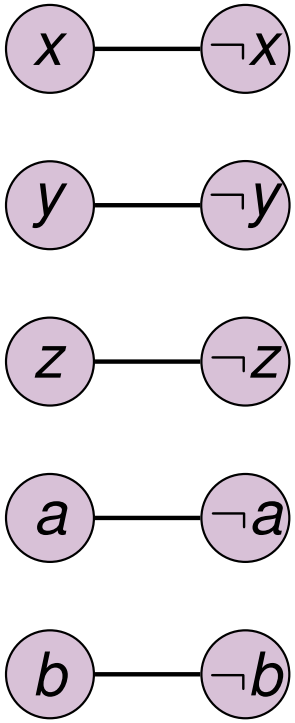

$k = 5 + \ldots$

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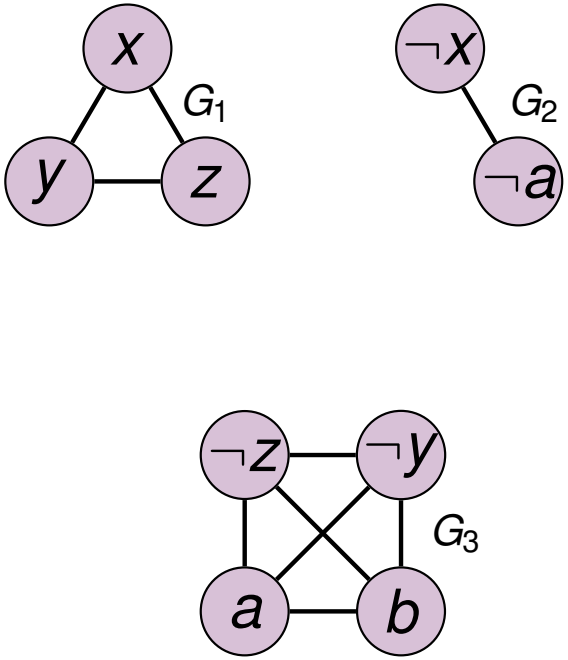

$k = \cdots + 2 + 1 + 3$

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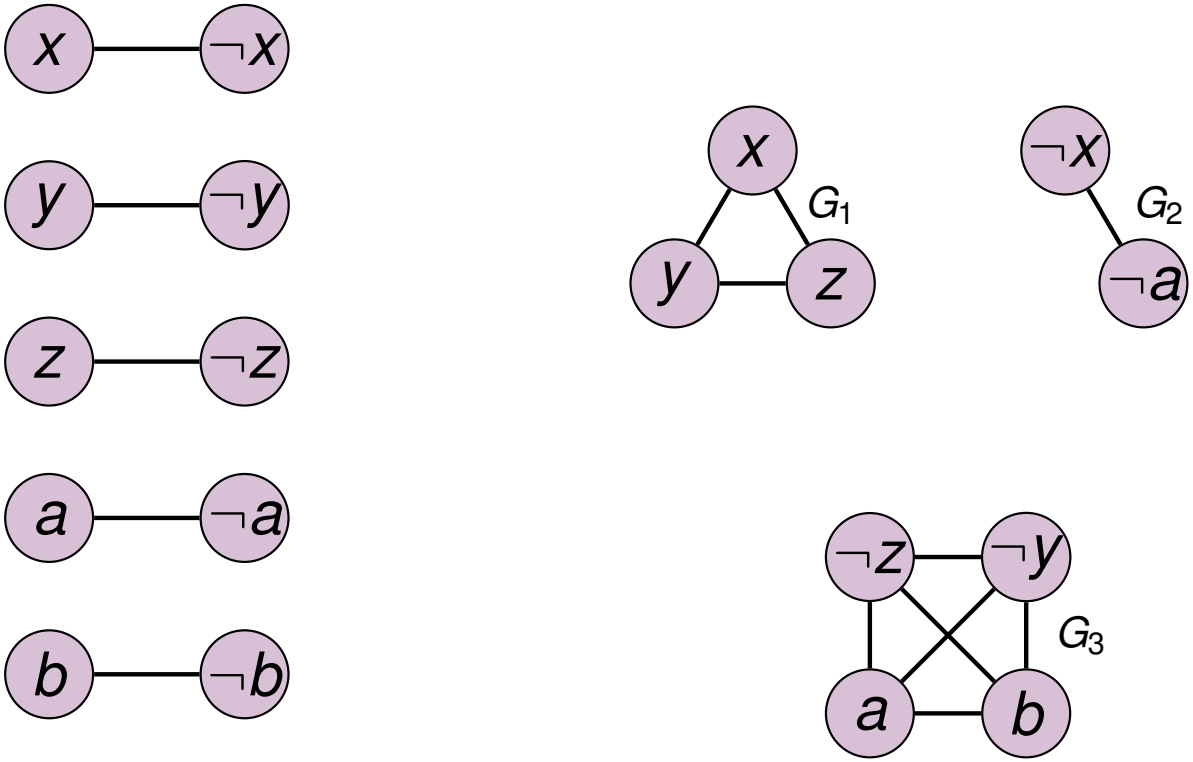

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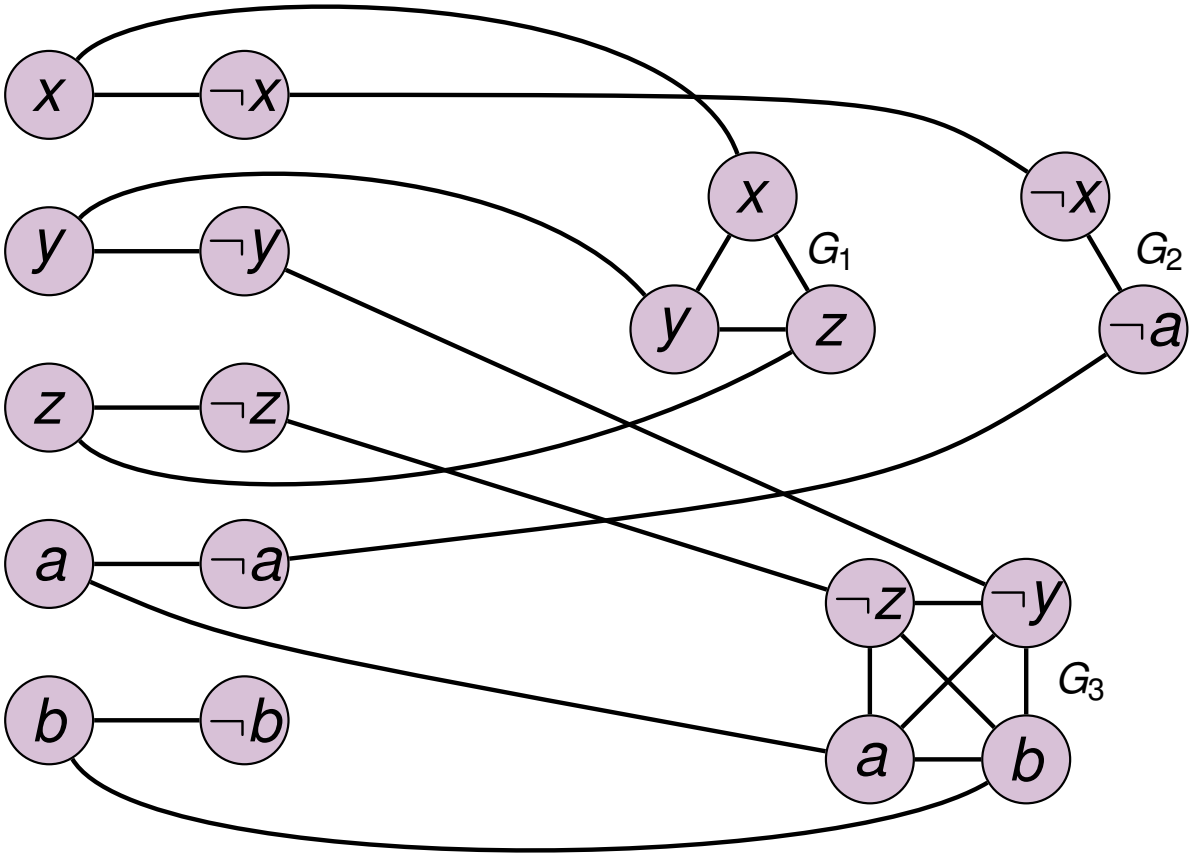

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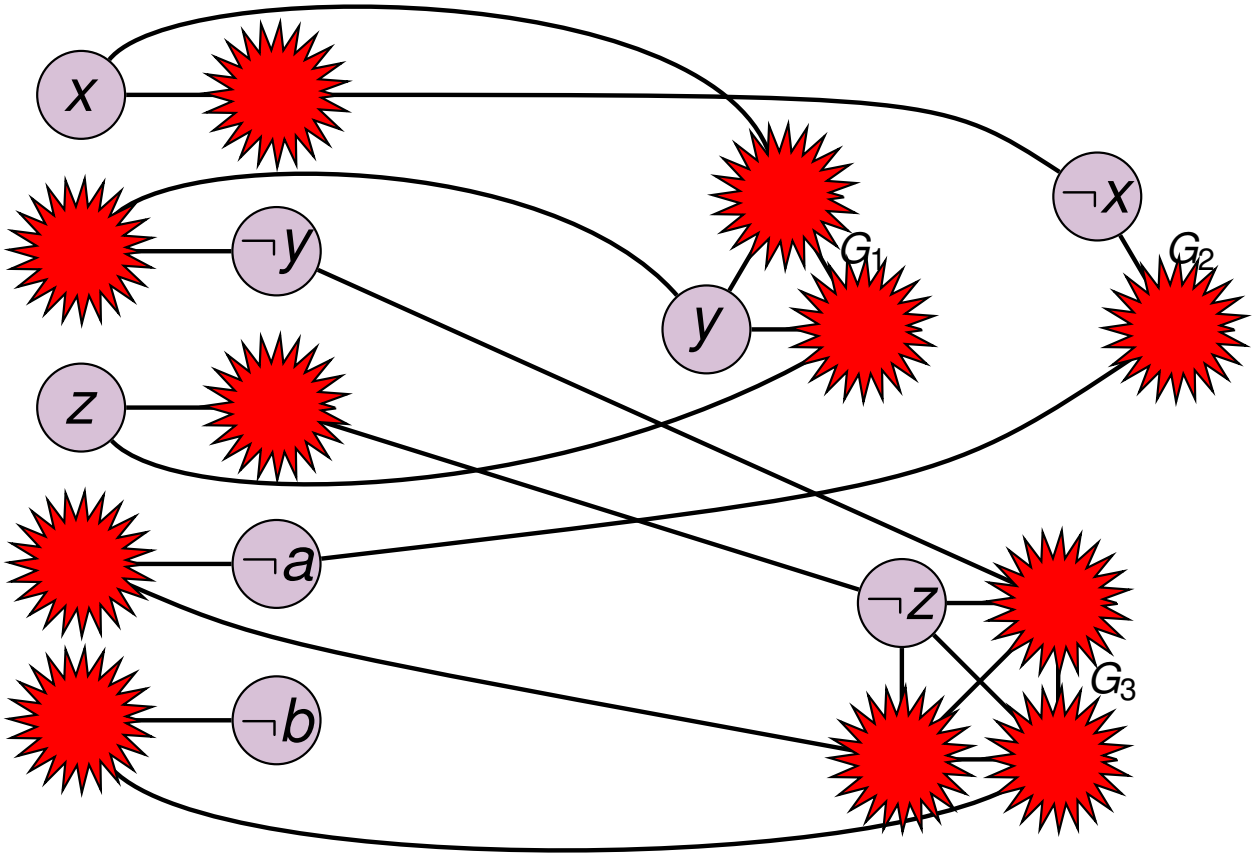

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Theorem

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- At least one of each pair (x^f, x^t) must be in the cover
- At least $n_j - 1$ vertices from each complete graph G_j must be in the cover
- If the formula is satisfiable, then choose the cover by choosing each literal assigned TRUE plus all but one vertex in each G_j. (Omit a vertex which is connected to a satisfied literal.)

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- Conversely, if a vertex cover exists, assign each Boolean variable according to whether x^t (TRUE) or x^f (FALSE) is in M.

  By the choice of k, there must be exactly one vertex in each clique which is not in M. This vertex must be adjacent to a literal-vertex in M, hence the clause is satisfied.

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Therefore, since SATISFIABILITY is **NP**-complete, it follows that:

Theorem

VERTEX COVER *is* **NP***-complete.*

NP-completeness of CLIQUE

CLIQUE
Instance: A finite graph $G = (V, E)$ and an integer k.
Question: Does G have a clique of size k?

- Recall (see Lecture 3) that there is a straightforward reduction from VERTEX COVER to CLIQUE: a set of vertices W is a vertex cover in G if and only if $V - W$ is a clique in the complement of G.

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- Recall (see Lecture 3) that there is a straightforward reduction from VERTEX COVER to CLIQUE: a set of vertices W is a vertex cover in G if and only if $V - W$ is a clique in the complement of G.
- Can we also find a reduction from SATISFIABILITY to CLIQUE?

Hitting Set

HITTING SET
Instance: Collection C of subsets of a set S and a positive integer k.
Question: Does S contain a *hitting set* for C of size k or less?
i.e.: a subset $S' \subseteq S$ with $|S'| \leq k$ such that S' contains at least one element form each subset from C?

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Which problem do you get then?

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Which problem do you get then? Vertex Cover !

Main idea:

- S is the set of vertices of a graph G
- the collection C of subsets of S is the set of edges of G

$\Rightarrow$ a hitting set of size k is a set of k vertices (from S) which "hit" (i.e. cover) all edges (in C)

3-SATISFIABILITY

To show that 3-SATISFIABILITY is **NP**-complete we reduce SATISFIABILITY to 3-SATISFIABILITY.

Proof idea:

- Replace every clause

$$C = x_1 \vee x_2 \vee \ldots \vee x_k$$

with $k > 3$ by:

$$C' = (x_1 \vee x_2 \vee y_1) \wedge (\neg y_1 \vee x_3 \vee y_2) \wedge \ldots \wedge (\neg y_{k-3} \vee x_{k-1} \vee x_k)$$

- C is satisfiable if and only if C' is, since at least one of the literals other than the y's must be TRUE .

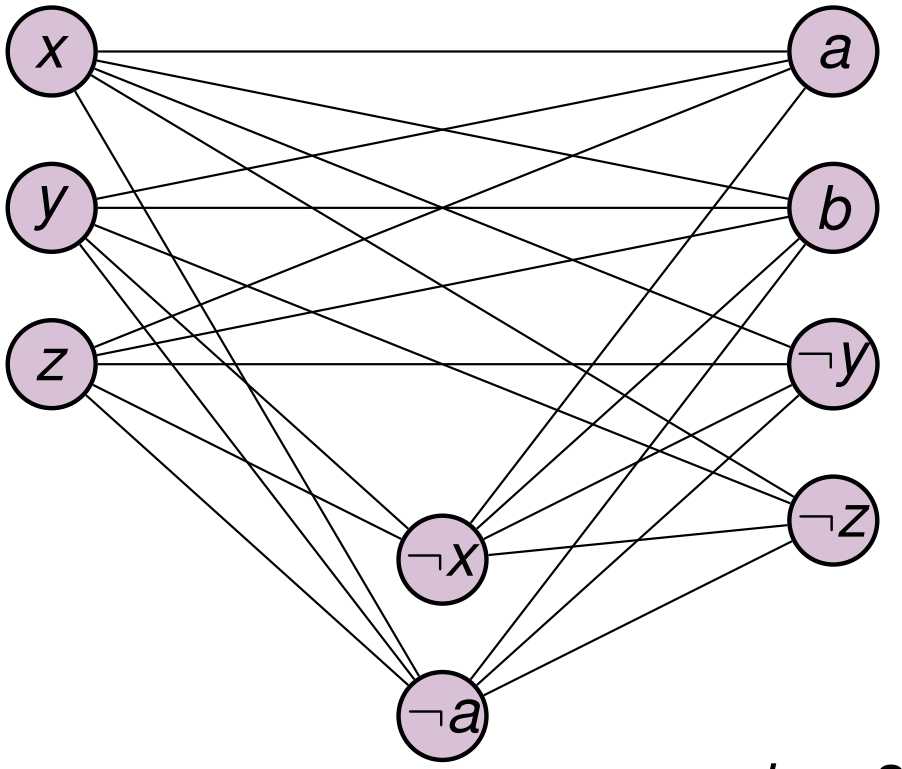
NP-completeness of CLIQUE

- The problem CLIQUE is in **NP** (the list of vertices in the clique is the certificate)
- To show that CLIQUE is **NP**-complete we shall reduce SATISFIABILITY to CLIQUE
- Given a formula f with clauses $C_1, C_2, \ldots, C_k$
  - For each literal x in clause C_j create a vertex
  - Put an edge between every pair of vertices from distinct clauses, unless they are incompatible (i.e. x and $\neg x$)

  This construction can be carried out in polynomial time
- The resulting graph has a clique of size k if and only if f is satisfiable (assign the value TRUE to every literal occurring in the clique)

Example construction

$$\underbrace{(x \vee y \vee z)}_{c_1} \wedge \underbrace{(\neg x \vee \neg a)}_{c_2} \wedge \underbrace{(a \vee b \vee \neg y \vee \neg z)}_{c_3}$$



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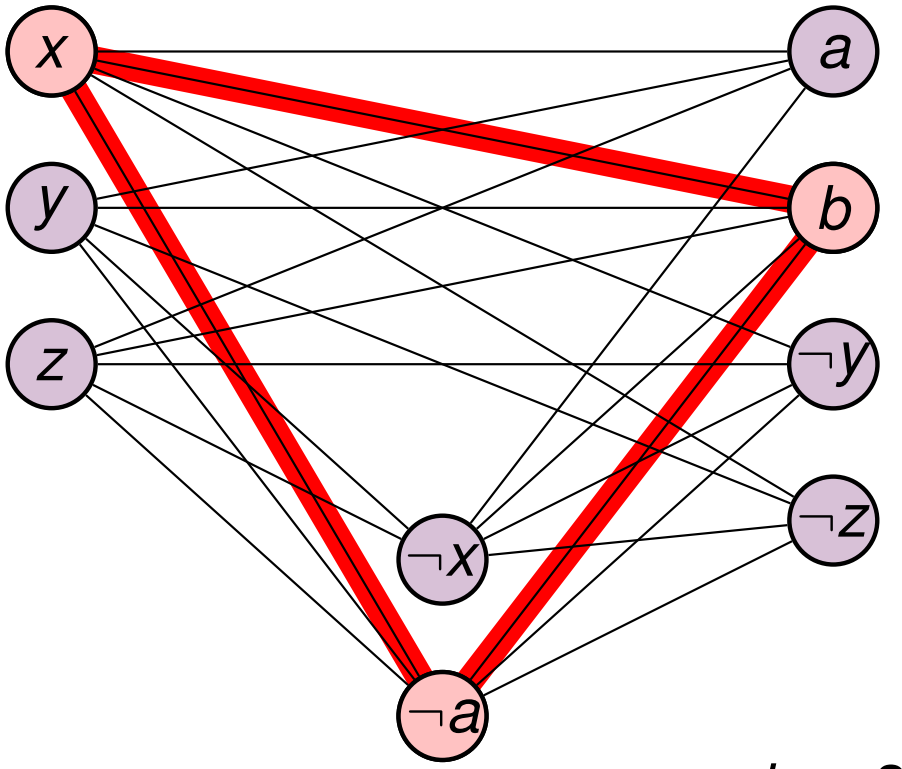

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Additional slides

NP-completeness of SUBSET SUM

SUBSET SUM
Instance: A set of positive integers $S = \{n_1, n_2, \ldots, n_k\}$, and a "target" integer t.
Question: Is there a subset $T \subseteq S$, such that $\sum_{i \in T} n_i = t$?

SUBSET SUM is in **NP** (the set T is the certificate)

To show that SUBSET SUM is **NP**-complete we will reduce SATISFIABILITY to SUBSET SUM.

NP-completeness of SUBSET SUM

Given a formula f with clauses $C_1, C_2, \ldots, C_k$

- For each variable x create two appropriate integers $\{n_{xt}, n_{xf}\}$
- For each clause C_j of length l_j, create $l_j - 1$ integers $\{m_{j1}, m_{j2}, \ldots\}$
- Choose the "target" integer T such that T must contain:
  - exactly one of each pair $\{n_{xt}, n_{xf}\}$

  $\Rightarrow$ this indicates that every variable is either TRUE or FALSE
  - at least one from each tuple $\{m_{j1}, m_{j2}, \ldots\}$

  $\Rightarrow$ this indicates that every clause is satisfiable

This construction can be carried out in polynomial time.

However, exact details about how we compute the integers $\{n_{xt}, n_{xf}\}$, $\{m_{j1}, m_{j2}, \ldots\}$, and t must be specified.

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- T includes exactly one of $\{n_{xt}, n_{xf}\}$ for each variable x; use this to decide whether each variable is TRUE or FALSE
  - i.e. the set T in the solution of the derived SUBSET SUM instance contains the information about which variables x are TRUE and which are FALSE in the initial SATISFIABILITY instance

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- for each clause C_i, T contains an integer of $\{n_{xt}, n_{xf}\}$ that represents a literal in C_i;

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- for each clause C_i, T contains an integer of $\{n_{xt}, n_{xf}\}$ that represents a literal in C_i;
- thus each clause (and so also the whole formula) is satisfied.

Example Construction

$$(x \vee y \vee z) \wedge (\neg x \vee \neg a) \wedge (a \vee b \vee \neg y \vee \neg z)$$

		x	y	z	a	b	C_1	C_2	C_3
n_{xt}	=	1	0	0	0	0	1	0	0
n_{xf}	=	1	0	0	0	0	0	1	0
n_{yt}	=		1	0	0	0	1	0	0
n_{yf}	=		1	0	0	0	0	0	1
n_{zt}	=			1	0	0	1	0	0
n_{zf}	=			1	0	0	0	0	1
n_{at}	=				1	0	0	0	1
n_{af}	=				1	0	0	1	0
n_{bt}	=					1	0	0	1
n_{bf}	=					1	0	0	0
m_{11}	=						1	0	0
m_{12}	=						1	0	0
m_{21}	=						0	1	0
m_{31}	=						0	0	1
m_{32}	=						0	0	1
m_{33}	=						0	0	1
t	=	1	1	1	1	1	3	2	4

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